

Chapter 6: Elementary Matrices

Definition: An $n \times n$ **elementary** matrix is a matrix obtained from the identity matrix by performing a single elementary row operation.

Example: Fill in the row operation that is performed on I_3 to get each elementary matrix, then perform the given matrix multiplication on an **arbitrary 3×3 matrix**. What do you observe?

$$\begin{aligned}
 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} &\xrightarrow{r_1 \leftrightarrow r_3} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} g & h & i \\ d & e & f \\ a & b & c \end{bmatrix} \\
 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} &\xrightarrow{r_3 \rightarrow 4r_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 4g & 4h & 4i \end{bmatrix} \\
 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} &\xrightarrow{r_1 \rightarrow r_1 + 3r_2} \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a+3d & b+3e & c+3f \\ d & e & f \\ g & h & i \end{bmatrix}
 \end{aligned}$$

Theorem: If an elementary row operation is performed on the $m \times n$ matrix A , then the result is the product EA , where E is the elementary matrix obtained by performing the same row operation on the $m \times m$ identity matrix I_m .

Fact: Every elementary matrix is invertible, and its inverse is an elementary matrix.

Example: Find the inverse of $\begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

$$\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Theorem: Suppose that $E_1, E_2, \dots, E_k$ are the elementary matrices that row reduce A to its reduced row echelon form, i.e. $E_k E_{k-1} \cdots E_2 E_1 A = RREF(A)$. Then:

1. A is invertible if and only if the reduced row echelon form of A is I_n .
2. When A is invertible, its inverse is $A^{-1} = E_k E_{k-1} \cdots E_2 E_1$.

Finding A^{-1}

Let A be a nonsingular $n \times n$ matrix. By the above theorem, $A^{-1} = E_k E_{k-1} \cdots E_2 E_1$. This gives us a method for finding A^{-1} : Perform elementary row operations $E_1, E_2, \dots, E_k$ on A until we get I_n , then $E_k E_{k-1} \cdots E_2 E_1$ equals A^{-1} .

A convenient way to track this computation is to write down the partitioned matrix $[A \mid I_n]$. Then applying the elementary row operations we transform $[A \mid I_n]$ into $[I_n \mid A^{-1}]$.

Algorithm for computing A^{-1}

1. Set up the augmented matrix $[A \mid I_n]$.
2. Apply elementary row operations to $[A \mid I_n]$ to reduce it to $[I_n \mid A^{-1}]$.

If A is an $n \times n$ matrix, either A can be reduced to I_n by elementary row operations or it cannot. In the first case, the algorithm produces A^{-1} ; in the second case, A^{-1} does not exist.

Example: Find A^{-1} where $A = \begin{bmatrix} 1 & 1 & 4 \\ 2 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix}$.

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 4 & 1 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{r_2 \rightarrow r_2 - 2r_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 4 & 1 & 0 & 0 \\ 0 & 1 & -6 & -2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\begin{array}{l} \xrightarrow{r_2 \rightarrow r_2 + 6r_3} \\ \xrightarrow{r_1 \rightarrow r_1 - 4r_3} \end{array} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & -4 \\ 0 & 1 & 0 & -2 & 1 & 6 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{r_1 \rightarrow r_1 - r_2}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -1 & -10 \\ 0 & 1 & 0 & -2 & 1 & 6 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right], \text{ so } A^{-1} = \begin{bmatrix} 3 & -1 & -10 \\ -2 & 1 & 6 \\ 0 & 0 & 1 \end{bmatrix}$$

Box Of Facts/Invertibility Criteria

Let A be an $n \times n$ matrix. The following are equivalent:

1. A is nonsingular/invertible.
2. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
3. A is row equivalent to I_n .
4. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.
5. $\text{rank}(A) = n$.
6. A can be written as a product of elementary matrices.